Why the direct-minimization embedding state is an excited eigenstate

(ghost Gutzwiller for the t_c-Cederbaum model, $B_c = 3$)

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This note explains a numerical observation made when the solution of the direct minimization (`not_imp_gga_cederbaum.py`) was used as the starting point of the embedding fixed-point loop (`gga_cederbaum.py`) for the t_c-Cederbaum model of `cederbaum.tex`: the core embedding state Φ_c of the direct minimization is an *excited* eigenstate of the embedding Hamiltonian, which the loop (that takes the lowest state) cannot follow. Notation is that of `cederbaum.tex`.

1 Statement

"Excited" does *not* refer to the physical system: the direct-minimization point is the lowest-energy solution of the constrained problem (its energy agrees with the exact ground-state energy to $\sim 10^{-7}$). It refers to the embedding Hamiltonian $H^c_{\rm imp}$ built from the multipliers recovered at that point: Φ_c is an exact eigenstate of $H^c_{\rm imp}$, but not the lowest one.

2 The two procedures and what each requires

Both stationarize the same Lagrangian

$$\mathcal{L} = E + \sum_I 2\,\mathrm{Tr}\big[\lambda_I \big({}^I\Delta(\Phi_I) - \rho^{II}\big)\big] - \sum_I E^c_I \big(\langle\Phi_I|\Phi_I\rangle - 1\big), \tag{1}$$

with E of Eq. (E) of `cederbaum.tex` and multipliers λ_I (density constraint) and E^c_I (normalization).

Direct minimization minimizes E over Φ_I and ψ_{qp} subject to the constraints. At a constrained minimum, $\mathcal{L}$ is stationary in Φ_I, i.e.

$$H^I_{\rm imp}\,\Phi_I = E^c_I\,\Phi_I, \tag{2}$$

where, for the core,

$$H^c_{\rm imp} = H^{loc}_c + \sum_{a\sigma} h_a \big(d^\dagger_{a\sigma} c_\sigma + \text{h.c.}\big) + \sum_{ab\sigma} \lambda^c_{ab}\, d^\dagger_{a\sigma} d_{b\sigma} + n_c \Big[\alpha + 2\mathrm{Tr}\Xi - \sum_{ab\sigma} \Xi_{ab}\, d^\dagger_{a\sigma} d_{b\sigma}\Big], \qquad \lambda^c = \Lambda - \lambda, \tag{3}$$

(with h_a, Λ, α, Ξ the derivatives of E given in Sec. Direct minimization of `cederbaum.tex`). Equation (2) says that Φ_I is *an eigenstate* of $H^I_{\rm imp}$; it says nothing about which one.

The impurity loop *assumes* that Φ_I is the *lowest* eigenstate of H^I_{imp} at the current multipliers (it diagonalizes H^I_{imp} and takes the ground state). This is the saddle-point reading of $\mathcal{L}$: minimize over Φ_I at fixed multipliers, then adjust the multipliers to satisfy the constraints. It is equivalent to the constrained minimum when the problem has no duality gap (e.g. if it were convex).

3 Numerical check

At $V = 1$, $t_c = 0.5$, $B_c = 3$ ($\Delta = 1$, $E_c = 10$), with the multipliers recovered from the direct-minimization point:

- Φ_c is an exact eigenstate of H^c_{imp} (overlap$^2 = 1.000000$), as Eq. (2) requires.
- It is the 15th eigenstate (index 14): $\langle\Phi_c|H^c_{\text{imp}}|\Phi_c\rangle = -22.04$ against a lowest eigenvalue of -42.58.
- Φ_o and Φ_v are ground states of their embedding Hamiltonians.
- The bath levels of λ^c_c are -11.27, -2.23 and $+2.24$ (eigenvalues of the 3×3 matrix); the core has ${}^c\Delta$ eigenvalues 1.2×10^{-5}, 0.9979 and 1.0000.

4 Why it can be excited

1. **Stationarity is not minimality.** Stationarity of $\mathcal{L}$ in Φ gives Eq. (2) for *some* eigenstate. It is the *constraint*, not the energy of H_{imp}, that selects which one.
2. **The constraint fixes the bath occupations.** ${}^c\Delta = \mathbb{I} - \Delta^{bb}$ must equal ρ^{cc}, the local density matrix of the quasiparticle state. With one bath electron per spin, this fixes *which* bath orbital is filled: the one that corresponds to the empty quasiparticle orbital. The multipliers λ_I are then adjusted so that this state is stationary.
3. **That need not be the lowest state at the recovered multipliers.** At these multipliers the lowest state of H^c_{imp} would put the bath electron in the -11.27 bath level (and, with it, change the core occupation), which contradicts the constraint satisfied by Φ_c. The state that satisfies the constraint is a different, higher eigenstate.
4. **Non-convexity.** The constraint is quadratic in Φ and the energy functional is nonlinear in the quantities extracted from Φ (through $R_I = S^{-1}D$ and $\bar{\Theta}$), so the problem is not convex and a duality gap is possible: the minimum of $\mathcal{L}$ over Φ at the optimal multipliers is not the constrained minimum. This is why the saddle-point assumption of the loop can fail.

Points 1 and 2 follow directly from the equations; points 3 and 4 are my reading of the numerical result, not a proof.

5 Consequences for the impurity loop

Lowest-state selection. Started from the full direct-minimization state (`state0`), one loop iteration already moves away: the residual $\|\rho^{II} - (1 - \Delta^{bb}_I)\|$ is 1.4, and the loop collapses to $\nu_c \approx 0$ (shift $\varepsilon = 10^{-9}$).

Maximum-overlap (MOM-like) selection. In each embedding solve, the eigenstate with the largest overlap with the previous Φ_I is taken instead of the lowest one (`follow=True`; degenerate subspaces are projected). One step from the direct-minimization state then gives a residual of 7.6×10^{-3} (instead of 1.4), but it is still not a fixed point at $\varepsilon = 10^{-9}$, for two reasons: (i) a filled ghost orbital has ${}^c\Delta$ eigenvalue 1.0000, so $S_c \sim \sqrt{\varepsilon}$ and λ^c is ill conditioned: a change of ρ of 2×10^{-6} changes λ^c by ≈ 3; (ii) the followed state sits in a dense stretch of the embedding spectrum (level spacing $\lesssim 2 \times 10^{-3}$). After that the loop drifts to the $B_c = 1$-like state or to the trivial product state ($E = -22$, $R_c = 0$).

Dependence on the shift (300 iterations, started from the direct-minimization state):

shift ε	lowest-state selection	max-overlap selection
10^{-9}	no convergence	no convergence or product state
10^{-8}, 10^{-7}	no convergence	some mixings converge, to the product state ($R_c = 0$)
10^{-6}	no convergence	no convergence
10^{-5}	no convergence (residual 2.4)	**converges** (residual $\sim 10^{-13}$) to a nearby minimum, E higher by 1.0×10^{-4}

The shifts 10^{-4} and 10^{-3} were not completed. At $\varepsilon = 10^{-5}$ the loop point is itself a constrained local minimum (restarting the direct minimizer from it lowers E by $< 10^{-12}$), with the same single pole in G^h_{cc}.

Conclusion. The impurity loop reproduces a solution of this kind only with maximum-overlap state selection and a large regularization shift; and then it finds a different, higher local minimum. The direct minimization needs no choice of eigenstate and is the more reliable solver.

Scope. The numbers are for $V = 1$, $t_c = 0.5$, $B_c = 3$ (shift sweep: $V = 1$, $t_c = 0.5$). The explanation in Sec. 3, points 3–4, is an interpretation of those numbers.